# How To Speak Tensors

## (with 4 interesting examples)

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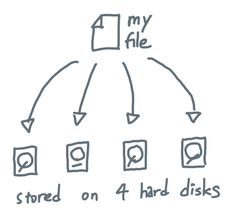
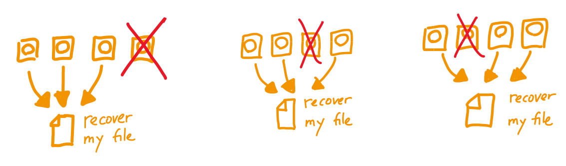
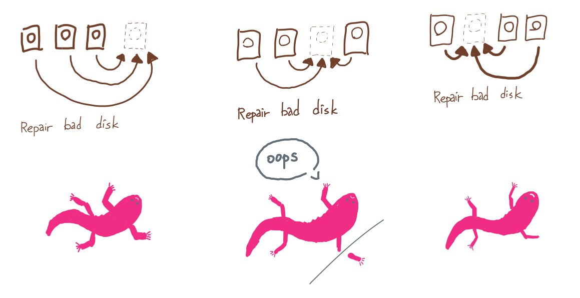
# Cloud Storage

Requirement




"Regenerating Code"




## Use Tensor

Pick a finite field $\mathbb{F}$, $|\mathbb{F}| \geq 4$

Let $V := \mathbb{F}^3$

Encode my file as a map

$\varphi: \ S^2V \longrightarrow \mathbb{F}$

where $S^2V := V \otimes V \big/ \{a \otimes b \equiv b \otimes a\}$ $\begin{smallmatrix} a \in V \\ b \in V \end{smallmatrix}$

Elements of $S^2V$ are linear combinations of $a \odot b$

Storage: #1 stores $\varphi(u_1 \odot V)$ i.e., $\varphi(u_1 \odot b)$ $\forall b \in V$ i.e., $\varphi \restriction u_1 \odot V$

#2 stores $\varphi(u_2 \odot V)$

#3 stores $\varphi(u_3 \odot V)$

#4 stores $\varphi(u_4 \odot V)$

File recovery: If ~~#4~~

$u_1 \odot V + u_2 \odot V + u_3 \odot V$

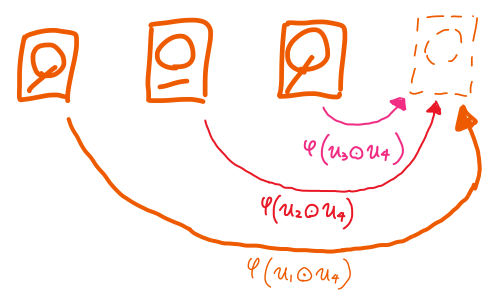
$= \operatorname{span}(u_1, u_2, u_3) \odot V$

$= V \odot V$

$= S^2V$

$\Rightarrow$ we have full $\varphi$

## Disk Repairment




$\operatorname{span}(u_1 \odot u_4, \ u_2 \odot u_4, \ u_3 \odot u_4)$

$= \operatorname{span}(u_1, u_2, u_3) \odot u_4$

$= V \odot u_4$

$= u_4 \odot V$

$\Rightarrow$ #4 has $\varphi(u_4 \odot V)$

[10] arXiv:2006.16998 [pdf, ps, other]
**Multilinear Algebra for Minimum Storage Regenerating Codes**
Iwan Duursma, Hsin-Po Wang
Comments: 36 pages, 9 figures, 3 tables
Subjects: **Information Theory (cs.IT)**; Commutative Algebra (math.AC)

[11] arXiv:2006.08911 [pdf, ps, other]
**Multilinear Algebra for Distributed Storage**
Iwan Duursma, Xiao Li, Hsin-Po Wang
Comments: 33 pages, 6 figures, 1 table
Subjects: **Information Theory (cs.IT)**; Commutative Algebra (math.AC)

# Matrix Multiplication

Want to compute

$$\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

Addition & subtraction is free.

How many multiplications do we need?

Answer: 8, because

$c_{11} = a_{11} \cdot b_{11} + a_{12} \cdot b_{21}$

$c_{12} = a_{11} \cdot b_{12} + a_{12} \cdot b_{22}$

$c_{21} = a_{21} \cdot b_{11} + a_{22} \cdot b_{21}$

$c_{22} = a_{21} \cdot b_{12} + a_{22} \cdot b_{22}$

But actually 7 is enough

$M_1 = (a_{11} + a_{22}) \cdot (b_{11} + b_{22})$

$M_2 = (a_{21} + a_{22}) \cdot b_{11}$

$M_3 = a_{11} \cdot (b_{12} - b_{22})$

$M_4 = a_{22} \cdot (b_{21} - b_{11})$

$M_5 = (a_{11} + a_{12}) \cdot b_{22}$

$M_6 = (a_{21} - a_{11}) \cdot (b_{11} + b_{12})$

$M_7 = (a_{12} - a_{22}) \cdot (b_{21} + b_{22})$

## Because

$c_{11} = M_1 + M_4 - M_5 + M_7$

$c_{12} = M_3 + M_5$

$c_{21} = M_2 + M_4$

$c_{22} = M_1 - M_2 + M_3 + M_6$

## "Strassen Alg"

Define $\langle n, n, n \rangle :=$ the cost of multiplying $n \times n$ with $n \times n$

Question: find $\underset{\text{matrix multiplication exponent}}{\omega}$ s.t.

$\langle n, n, n \rangle \approx n^{\omega}$

Exercise: $\langle 5, 5, 5 \rangle \leq$

$\langle 2,2,2 \rangle + \langle 2,3,2 \rangle + \langle 2,2,3 \rangle + \langle 2,3,3 \rangle +$

$\langle 3,2,2 \rangle + \langle 3,3,2 \rangle + \langle 3,2,3 \rangle + \langle 3,3,3 \rangle$

$= \langle 2,2,2 \rangle + 3\langle 2,2,3 \rangle + 3\langle 2,3,3 \rangle + \langle 3,3,3 \rangle$

Hint $\boxplus \cdot \boxplus$

Lesson: knowing small $\langle p, q, r \rangle$ help us bound $\omega$ because

$\langle pqr, pqr, pqr \rangle = \langle p,q,r \rangle \cdot \langle q,r,p \rangle \cdot \langle r,p,q \rangle$

$= \langle p,q,r \rangle^3$

What if we know $\langle p,q,r \rangle + \langle a,b,c \rangle$ ?

$[p,q,r]$ is a tensor

$\sum_{i=1}^{p} \sum_{j=1}^{q} \sum_{k=1}^{r} e_{ij} \otimes e_{jk} \otimes e_{ik} \in \mathbb{F}^{p \times q} \otimes \mathbb{F}^{q \times r} \otimes \mathbb{F}^{p \times r}$

Idea: $\left([p,q,r] \oplus [q,r,p] \oplus [r,p,q] \oplus [a,b,c] \oplus [b,c,a] \oplus [c,a,b]\right)^{\otimes m}$

$$= \bigoplus_{m_1, m_2, m_3, m_4, m_5, m_6} \binom{m}{m_1, m_2, m_3, m_4, m_5, m_6} [p,q,r]^{\otimes m_1} \otimes [q,r,p]^{\otimes m_2} \otimes \cdots \otimes [c,a,b]^{\otimes m_6}$$

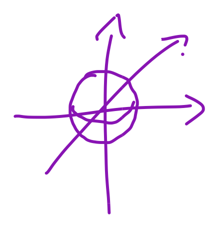
## Law of large number

$$\approx \binom{m}{\frac{m}{6} \ \frac{m}{6} \ \frac{m}{6} \ \frac{m}{6} \ \frac{m}{6} \ \frac{m}{6}} [pqrabc, pqrabc, pqrabc]^{\otimes \frac{m}{6}}$$

Corollary: $(pqr)^{\omega} + (abc)^{\omega} \leq \langle p,q,r \rangle + \langle a,b,c \rangle$

# Quantum & Testability

Bloch Sphere



describes the spin of an electron.

Spin up $\uparrow$ $|\uparrow\rangle$   Spin down $\downarrow$ $|\downarrow\rangle$   Spin plus $\rightarrow$ $|+\rangle$   Spin minus $\leftarrow$ $|-\rangle$

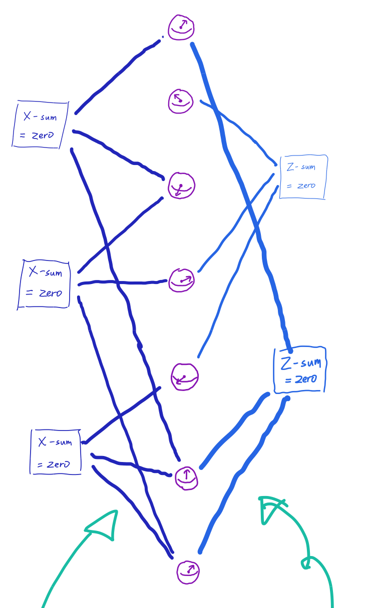
We store quantum information using spins

But there could be two types of errors

X-error = ↻ 180°    Z-error = ↺ 180°

A quantum fact: other errors are just combination of X & Z-errors

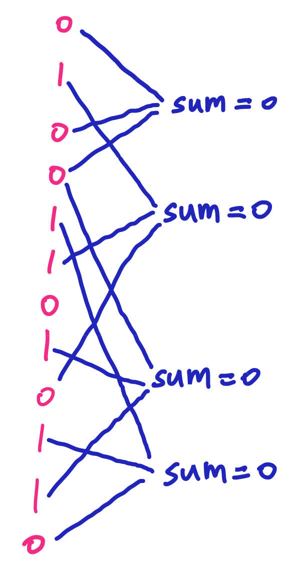
To protect quantum information, use X-checks & Z-checks




This is a bipartite graph

This is another bipartite graph

This is analogous to classical parity-check codes




bipartite graph $\Rightarrow$ biadjacency matrix $\Rightarrow$ linear map

$$\mathbb{F}_2^{a} \xrightarrow{f_X} \mathbb{F}_2^{n} \xrightarrow{f_Z} \mathbb{F}_2^{b} \qquad (*)$$

A quantum fact: we need $f_Z \circ f_X = 0$

So $(*)$ is a chain complex

How to construct chain complex?

Hastings–Haah–Donnell [2009-03921]:

use tensor product

$$\begin{tikzcd}[column sep=small]
& A \arrow[dl, "f"'] & C \arrow[dr, "g"] & \\
B & & & D
\end{tikzcd}$$

$$\begin{tikzcd}
& A \otimes C \arrow[dl, "f \otimes \mathrm{Id}_C"'] \arrow[dr, "\mathrm{Id}_A \otimes g"] & \\
B \otimes C \arrow[dr, "\mathrm{Id}_B \otimes g"'] & & A \otimes D \arrow[dl, "f \otimes \mathrm{Id}_D"] \\
& B \otimes D &
\end{tikzcd}$$

$\Downarrow$

$B \otimes C \longrightarrow A \otimes C \oplus B \otimes D \longrightarrow A \otimes D$